

Trends and variability of water balance components over a tropical savanna and Eucalyptus forest in Australia

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ABSTRACT

In this paper, the long-term dynamics of water balance components in two different contrasting ecosystems in Australia were simulated with an ecohydrological model (WATER Vegetation Energy and Solute modelling (WAVES)) over the period 1950–2015. The selected two ecosystems are woodland savanna in Daly River and eucalyptus forest in Tumbarumba. The WAVES model was first manually calibrated and validated against soil water content measured by cosmic-ray probe and evapotranspiration measured with eddy flux techniques. The calibrated model was then used to simulate long-term water balance components with observed climate data at two sites. Analyzing the trends and variabilities of potential evapotranspiration and precipitation is used to interpret the climate change impacts on ecosystem water balance. The results showed that the WAVES model can accurately simulate soil water content and evapotranspiration at two study sites. Over the period of 1950–2015, annual evapotranspiration at both sites showed decreasing trends ($-1.988 \text{ mm year}^{-1}$ in Daly and $-0.381 \text{ mm year}^{-1}$ in Tumbarumba), whereas annual runoff in Daly increased significantly ($5.870 \text{ mm year}^{-1}$) and decreased in Tumbarumba ($-0.886 \text{ mm year}^{-1}$). It can be concluded that the annual runoff trends are consistent with the rainfall trends, whereas trends in annual evapotranspiration are influenced by both rainfall and potential evapotranspiration. The results can provide evidence for controlling the impacting factors for different ecosystems under climate change.

Key words: ecosystem, evapotranspiration, historical climate, soil water content, WAVES

HIGHLIGHTS

- This study highlights the importance of long-term climatic variability in evaluating water balance components at different vegetation ecosystems.
- The manually calibrated ecohydrological model is adopted in this paper.

INTRODUCTION

Climate change can influence the energy and water balance of the ecosystem resulting in changes in water yield and ecosystem services (Chen *et al.* 2016; Zhu *et al.* 2017). Therefore, it is necessary to study how the hydrological processes respond to climatic changes in various ecosystems with long historical data. It is considered that vegetation cover and land-use type are two main factors that affect the ecohydrological processes since they can change the water balance and water distribution among evapotranspiration (ET), runoff, soil water content and groundwater (Chen *et al.* 2016). For temperate forest, precipitation is the main controlling factor, but temperature and moisture are also important factors (Leuning *et al.* 2005; Kray *et al.* 2012). For the re-vegetation catchment area in Loess Plateau, different land-cover types can affect the water infiltration and drying front (Wang *et al.* 2013). On the contrary, soil water can also affect vegetation growth, variability and regeneration (Bogena *et al.* 2013). As a key factor in the hydrological cycle, soil moisture is mainly determined by the processes of infiltration, percolation, evaporation and root water uptake and is always used to evaluate the exchange of water and energy at soil surface.

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In this study, an ecohydrological model such as Water Vegetation Energy and Solute modelling (WAVES, see Zhang & Dawes 1998) was used to investigate the climate change impacts on water balance components for two different ecosystems. Zhang *et al.* (1996) used the WAVES model to simulate net radiation, ET and soil water content compared with the experimental data, the results showed that the simulated data agreed well with the experimental data and the model also has some limitations because of many soil and vegetation parameters. Zhang *et al.* (1999) used the WAVES model to test lucerne growth and groundwater uptake with the salinity and groundwater level changes, and the results also showed that the WAVES model can simulate well and can be used in the irrigated agricultural systems. In recent years, the WAVES model was used to study the impacts of climate change and CO₂ concentration increase on hydrological processes (Crosbie *et al.* 2010; Cheng *et al.* 2014a, 2014b). Crosbie *et al.* (2010) enlarged the WAVES model application scale from point to the basin after the study of future climate impacts on the groundwater recharge in the Murray–Darling Basin in Australia. The WAVES model was also used to simulate soil water moisture, water-use efficiency and crop yield (Kang *et al.* 2003; Tian *et al.* 2017; Ye *et al.* 2021). Kang *et al.* (2003) studied the temperature and rainfall change impacts on soil water content and crop yield under 10 climate scenarios in Taihang Mountain, and the results showed that the change of soil water content is in accordance with the change of vegetation productivity and variation of the Leaf Area Index (LAI) can decrease the temperature and rainfall impacts on soil water content to some extent. Tian *et al.* (2017) used the WAVES model to study the vegetation restoration effect on soil water content in the Yangjuangou Basin of Loess Plateau. Ye *et al.* (2021) employed the WAVES model to analyze water use in different vegetation land under future climate scenarios during the 2030s, 2050s and 2080s in the Xiong'an New area. The previous studies showed that the WAVES model has been successfully calibrated and used in the study on the variation of soil water content, groundwater and ET, especially in Australia and China.

This study selected two different Australian typical ecosystems to analyze the trends and variability of water balance components with long-time series climate data. One is the tropical savanna (located in Daly River), which is the dominant ecosystem in north Australia and largely influenced by land management practices such as animal husbandry and fire (Leuning *et al.* 2005), and it covers almost 25% of the Australian landmass with the area of 2 million km² (Beringer *et al.* 2007). Another one is Eucalyptus forest (located in Tumbarumba), which comprises 75% of the total Australian native forest with an area of 0.92 million km² in Australia. The Eucalyptus forest grows in the large rainfall area and has high productive and economical valuables (Leuning *et al.* 2005). Meanwhile, the two ecosystems have completely different climate types, Daly River has the tropical climate and Tumbarumba has an oceanic climate. The two study sites have been established, both Australian National Flux Network (OzFlux) and Cosmic-Ray Probe (CRP), which can provide the energy and soil moisture data to calibrate the WAVES model. OzFlux is a national ecosystem research network with eddy covariance flux towers to provide information on the exchanges of water vapor and energy between ecosystem and the atmosphere at a range of time and space scales (Glenn *et al.* 2011). CRP is a new promising method to measure soil water content at the field or small catchment scale (Zreda *et al.* 2012; Zhu *et al.* 2017). The effective depth of CRP is influenced by soil water content (~75 cm for dry soils and ~12 cm for wet soils) (Bogena *et al.* 2013). When the soil moisture is higher in the soil, the effective depth is shallower and vice versa.

This study has two objectives: (1) to calibrate and test the WAVES model against measured soil water content and ET measurements from two different ecosystems in Australia and (2) to analyze annual trends and variability of the water balance components of the ecosystems over the past 60 years as simulated by the WAVES model. The study can provide insight into key factors that control the water balance dynamics of the Australian ecosystems and their potential responses to future climate change.

STUDY AREA AND DATA

The study sites are the Daly River Uncleared flux tower and the Tumbarumba flux tower site (Figure 1), representing two different Australian ecosystems.

The Daly River Uncleared flux tower site (AU-DaS) is located in the Douglas River Daly River Esplanade Conservation area (14.16°S, 131.39°E), approximately 60 km south west of Pine Creek, Northern Territory. The vegetation cover is classified as a woodland savanna with an average canopy height of 16.4 m. The upper storey species are the evergreen *Eucalyptus tetrodonta* and *Corymbia latifolia*, whereas the understorey is covered by perennial grasses (Prior & Eamus 1999; Cernusak *et al.* 2011). The mean annual temperature is 27.4 °C and the mean annual rainfall is about 1,170 mm (<http://www.ozflux.org.au/>). The soil type is characterized as Haplic Mesotrophic Red Kandosol.

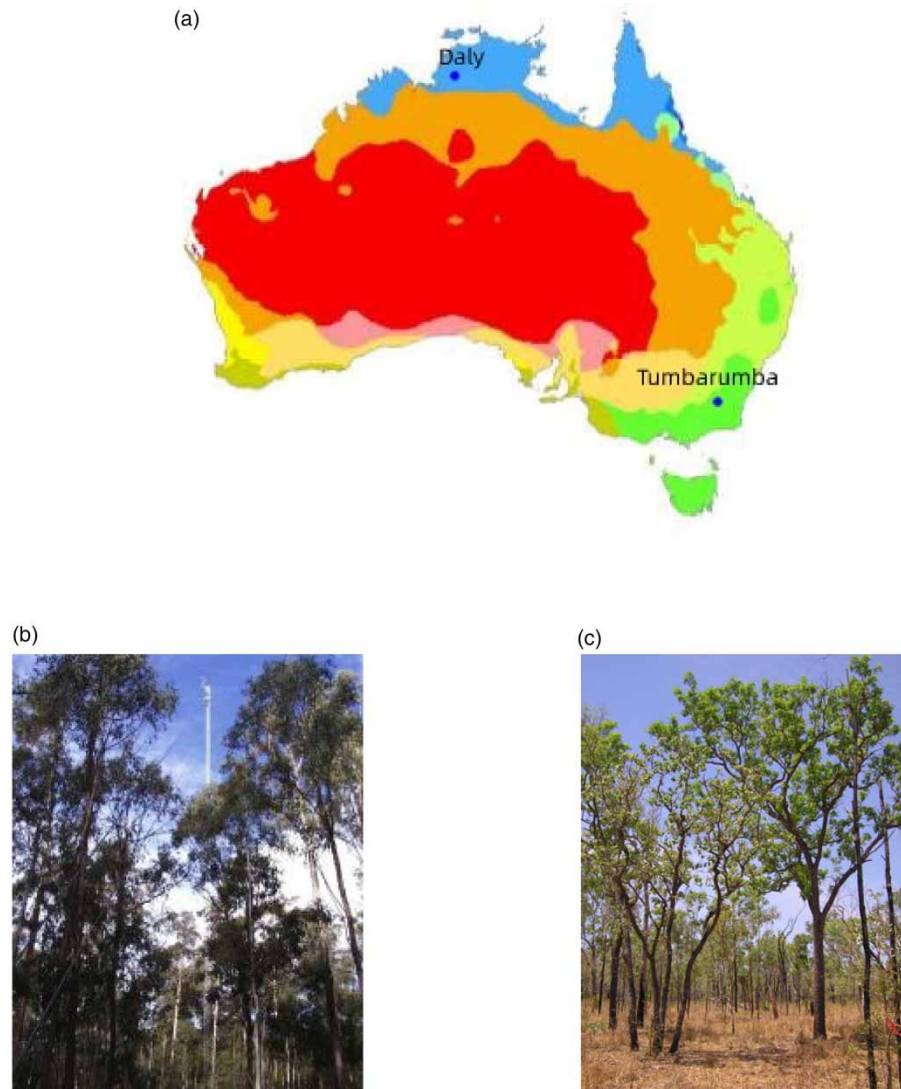


Figure 1 | Locations of study sites with climate map (a) and photos of Tumbarumba (b) and Daly (c) in Australia.

The Tumbarumba flux station (AU-Tum) is located in the Bago State forest in southeastern New South Wales (35.66°S, 148.15°E). The vegetation cover is wet sclerophyll forest with an average canopy height of 40 m. The upper storey species are *Eucalyptus delegatensis* and *Eucalyptus dalrymplean*, and the understorey is covered by sparse shrubs and full grasses (Leuning *et al.* 2005). The mean annual temperature is 9.3 °C, and the mean annual precipitation is about 1,000 mm (<http://www.ozflux.org.au>). The soil type is classified as an acidic, eutrophic red dermosol (Hawdon *et al.* 2014).

The soil water contents at both sites were measured using CRPs and obtained from CosmOz (<http://cosmoz.csiro.au>). The CRPs have been calibrated and tested by Hawdon *et al.* (2014) with a generalized calibration function to relate neutron counts with soil moisture content. The soil water contents were logged at hourly time intervals and averaged to daily values at the mean effective depth. ET for two sites was measured using the eddy covariance technique and obtained from OzFlux (<http://www.ozflux.org.au/>). The ET measurements were carried out at 30-min intervals and averaged to daily values with an energy balance closure approach (Leuning *et al.* 2008). At the Daly site, the soil moisture and ET data covered the period of 7 June 2011–22 July 2013. At the Tumbarumba site, the soil moisture and ET data covered the period of 3 April 2011–31 December 2011.

Daily values of maximum and minimum temperatures, vapor pressure deficit and total solar radiation were obtained from the 'SILO Data Drill' (<https://www.longpaddock.qld.gov.au/silo/>) over the periods of the soil moisture and ET

measurements. Rainfall data over these periods were obtained from CosmOz and shown in Figure 2. To investigate the long-term trends and variability of water balance components for these two ecosystems, daily values of meteorological variables over the period of 1950–2015 were obtained from the ‘SILO Data Drill’.

METHODS

WAVES model

The WAVES model is an ecohydrological model to predict the energy and water balance within the soil, atmosphere and vegetation system on a daily time step (Zhang & Dawes 1998).

The WAVES model can simulate all the water balance components in the hydrological processes, in particular rainfall infiltration, overland flow, soil evaporation and plant transpiration, soil moisture distribution, drainage and water table interactions. Soil water movement in both the saturated and unsaturated zones is simulated with a fully implicit finite-difference numerical solution of the Richards equation (Dawes & Short 1993). A full description of the solution to the Richards' equation can be found in Dawes & Short (1993). The assumption of Richards' equation in WAVES is that the soil is incompressible, non-hysteretic and isotropic, and the soil water flow is via the soil matrix only. Overland flow can be generated when positive soil water potential is generated at the soil surface. This can occur when the rainfall rate exceeds the conductivity and absorptivity capacity of the soil, or when rain falls on a saturated surface. Lateral subsurface flow can be generated via a perched water table at a soil layer boundary and is simulated by Darcy's law if the non-zero slope is specified. Groundwater is an alternative lower boundary condition, where the water table depth can be specified and may change daily within the soil profile, and water can be transferred to or from the soil column depending on internal conditions. ET is estimated using the Penman–Monteith equation with air temperature, vapor pressure deficit and solar radiation as inputs. A more detailed description of WAVES is provided in Zhang & Dawes (1998).

To solve the Richards equation, the analytical soil model of Broadbridge and White (BW) is employed to describe the relationships among soil water potential, volumetric water content and hydraulic conductivity (Zhang & Dawes 1998). The BW model can realistically represent a comprehensive range of soil moisture characteristics, from a highly nonlinear associated with a well-developed capillary fringe to a weakly nonlinear associated with highly structured soil and macropores.

The WAVES model has been successfully used to simulate vegetation water-use and carbon processes under different conditions (Wang *et al.* 2013; Chen *et al.* 2016) and quantify water budget with elevated CO₂ (Cheng *et al.* 2014a, 2014b). The

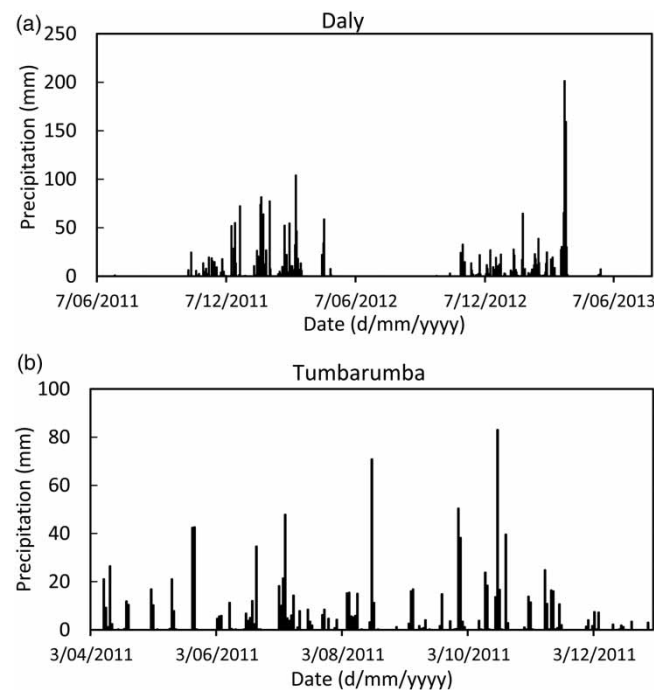


Figure 2 | Daily precipitation measured in Daly and Tumbarumba during the calibration period.

advantages of the WAVES model are the following: (1) accurately simulating the soil water dynamics under saturated and unsaturated soil and (2) accurately simulating the canopy transpiration due to the linkage between the hydrological process and vegetation growth at plot scales (Cheng *et al.* 2014b). The disadvantages include the following: (1) it cannot simulate the smaller time-scaled phenomena since it is a daily time-step model and (2) it cannot simulate plant phenology and select dynamic vegetation growth since the crop yield is decided by the harvest index of non-yielding and perennial plants (Dawes *et al.* 1998).

Model evaluation and application

Model calibration

In this study, the WAVES model was manually calibrated against soil moisture and ET measurements and the purpose of the model calibration was to obtain an optimal parameter set so that we can use the model to investigate the long-term trends and variability of water balance components for the selected ecosystems.

The initial soil hydraulic parameters for each soil type were set according to the typical values recommended by Dawes *et al.* (1998) for the BW model. The BW model has five parameters, including saturated hydraulic conductivity, saturated water content, air-dry water content, soil capillary length and soil water structure parameter. In this study, three soil layers in Daly and one soil layer in Tumbumba were considered. The simulation of vegetation growth in WAVES needs 26 parameters, which try to reflect the rate-based processes for vegetation growth, death and water allocation responding to different environmental conditions (Dawes *et al.* 1998). Before model calibration, the initial vegetation data were selected from the recommended values for eucalyptus trees and mixed perennial pastures for the understorey habitats contained in the WAVES manual (Dawes *et al.* 1998).

Since the initial values may have significant boundary effects on the simulated results at the beginning of the study period which may last several months, thus, the initial values of soil and vegetation must be given properly. The initial values of soil moisture are only related to a few parameters with explicit physical meanings and bounded ranges. Therefore, a trial and error method is used and a complex optimization method is not considered to minimize the differences between simulated and measured soil moisture and ET. The key parameters that were calibrated include soil hydraulic properties, maximum carbon assimilation rate, rainfall interception coefficient, optimal temperature for plant growth and specific leaf area (Table 1). All values listed in Table 1 are assumed to be time-invariant because the changes in biophysical parameters are scale-dependent in both magnitude and variability.

Model performance evaluation

In this paper, statistical indices are used to evaluate the accuracy of the WAVES model calibration, based on the comparisons of soil water content and ET between the observed data and simulation results. The indices include: (1) mean absolute error (MAE), (2) relative mean square error (RMSE) and (3) Nash–Sutcliffe efficiency (NSE, Nash & Sutcliffe 1970), which can be described as follows:

$$\text{MAE} = \frac{\sum_{i=1}^n |P_i - O_i|}{n} \quad (1)$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (P_i - O_i)^2}{n}} \quad (2)$$

$$\text{NSE} = 1 - \frac{\sum_{i=1}^n (P_i - O_i)^2}{\sum_{i=1}^n (O_i - \bar{O})^2} \quad (3)$$

where P_i is the simulated value of soil water content or ET by WAVES; O_i is the measured value of soil water content by CosmOz or ET by OzFlux; $\bar{O}$ is the average measured value of the sample sequence number; i is the simulated sample sequence number, $i = 1, 2, \dots, n$; and n is the total number of the data pairs.

Model application

Once the WAVES model was calibrated against the soil water content and ET measurements, the model was run using the optimized model parameter values (Table 1) with historical climatic data over the period of 1950–2015. This enables the examination of the effects of climate variability on soil water content, ET and runoff.

Table 1 | Values and units of key physiological parameters of the WAVES model in Daly and Tumbarumba

Parameter (unit)	Value			
	Daly		Tumbarumba	
	Upper storey	Understorey	Upper storey	Understorey
1 minus albedo of the canopy (–)	0.85	0.90	0.85	0.90
1 minus albedo of the soil (–)	0.90	0.90	0.90	0.85
Rainfall interception coefficient (m LAI ^{–1} d ^{–1})	0.0001	0.0005	0.0001	0.0001
Light interception coefficient (–)	–0.5000	–0.6500	–0.5000	–0.6500
Maximum carbon assimilation rate (kg C m ^{–2} d ^{–1})	0.0300	0.0100	0.0100	0.015
Slope parameter for the stomatal conductance model (–)	1.0	1.0	1.0	1.0
Maximum plant-available soil water potential (m)	–150	–150	–400	–150
IRM weighting of water relative to light (–)	2.5000	2.1000	2.5000	2.1000
IRM weighting of nutrients relative to light (–)	0.5000	0.5000	0.5000	0.5000
Temperature when growth rate is half optimum (°C)	20	10	15	10
Temperature when growth rate is optimum (°C)	25	20	25	20
Saturation light intensity (μmoles m ^{–2} d ^{–1})	1,500	1,200	1,500	1,200
Specific leaf area (m ² kg ^{–1})	10	12	10	12
Leaf respiration coefficient (kg C kg C ^{–1})	0.0015	0.0001	0.0002	0.0015
Stem respiration coefficient (kg C kg C ^{–1})	0.0001	–1.0000	0.0002	–1.0000
Root respiration coefficient (kg C kg C ^{–1})	0.0001	0.0001	0.0001	0.0015
Leaf morality rate (proportion leaf C d ^{–1})	0.0001	0.0050	0.0007	0.0050
Aerodynamic resistance (s m ^{–1})	20	30	20	30

Trend analysis

Trend analysis is used to determine whether the measured or calculated values of the climate parameters and water balance components increase or decrease during a time period. In this study, the Mann–Kendall (MK) test and Sen's slope estimator were used to identify the trends of historical climate and water balance components. Both methods have been widely used in hydro-meteorological time series. A brief description of the statistical methods is as follows.

The MK test is a non-parametric test to analyze the trends of climatologic and hydrologic time series. It does not need to accord with any particular distribution and has low sensitivity to the abrupt breaks owing to the inhomogeneous time series. The MK test statistic (S) (Mann 1945; Kendall 1975) is defined as follows:

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^n \text{sgn}(x_j - x_k) \quad (4)$$

where n is the number of data points, x_k and x_j are the data values in time series k and j ($j > k$), respectively, and $\text{sgn}(x_j - x_k)$ is the sign function as follows:

$$\text{sgn}(x_j - x_i) = \begin{cases} 1, & \text{if } (x_j - x_k) > 0 \\ 0, & \text{if } (x_j - x_k) = 0 \\ -1, & \text{if } (x_j - x_k) < 0 \end{cases} \quad (5)$$

The MK test has two parameters that are of importance to trend detection. One is the significance level, which indicates the strength of the estimated trend, and the other one is the slope magnitude estimate, which indicates the direction and the rate of change. Under the hypothesis that the data are independent and have the same distribution, the mean value of S is zero and

the variance of S in Equation (4) can be calculated as follows:

$$\text{var}(S) = \frac{n(n-1)(2n+5)}{18} \quad (6)$$

The normal Z -test statistic is calculated as follows:

$$Z = \begin{cases} (S-1)/\sqrt{\text{var}(S)} & \text{if } S > 0 \\ 0 & \text{if } S = 0 \\ (S+1)/\sqrt{\text{var}(S)} & \text{if } S < 0 \end{cases} \quad (7)$$

A positive value of Z indicates an increasing trend, while a negative value of Z represents a decreasing trend. For the two sides test, the null hypothesis is rejected at a significance level of α if $|Z| > Z_{(1-\alpha/2)}$, where $Z_{(1-\alpha/2)}$ is the value of the standard normal distribution with a probability of exceedance of $\alpha/2$.

If a linear trend is presented, the trend magnitude or the slope (change per unit time) Q is estimated by Sen's non-parametric method (Salmi *et al.* 2002). The equation can be expressed as follows:

$$f(t) = Qt + B \quad (8)$$

where Q is the slope and B is a constant.

$$Q = \text{Median} \left[\frac{x_j - x_k}{j - k} \right] \quad (9)$$

for all $k < j$, where $1 < k < j < n$. In other words, the slope estimator Q is the median of all possible combinations of pairs for the whole data set. In this study, the confidence intervals were calculated at three different confidence levels ($\alpha = 0.001$, $\alpha = 0.01$ and $\alpha = 0.05$).

RESULTS AND DISCUSSION

Model calibration

Soil water content

The simulated and measured soil water contents in Daly and Tumbarumba are presented in Figure 3, and statistics about model performance are listed in Table 2.

Figure 3(a) and 3(c) shows that the simulated soil water contents are consistent with the observed soil moisture contents for both Daly and Tumbarumba. Soil water content in Daly exhibited strong seasonal variation in response to rainfall variation (Figure 2(a)). However, the soil moisture contents in Tumbarumba did not show large variation due to the short period of soil moisture measurement. The soil water content in Daly ranged from 0.01 to 0.27 $\text{cm}^{-3} \text{cm}^{-3}$ from November to May, and at a relatively stable value (around 0.02 $\text{cm}^{-3} \text{cm}^{-3}$) from June to October. Soil water contents in Tumbarumba did not show strong seasonal change and varied between 0.37 and 0.79 $\text{cm}^{-3} \text{cm}^{-3}$. Examinations of Figures 2 and 3 reveal that the variations of soil water content are consistent with rainfall variability. When rainfall occurs, soil water content increases and it starts to decrease in dry seasons. The reason why the soil water content in Tumbarumba is too much higher than that in Daly is that the litter layer occurs in Tumbarumba and the bulk density is lower in Tumbarumba (0.95 g cm^{-3}) than in Daly (1.48 g cm^{-3}) (Hawdon *et al.* 2014). Meanwhile, the measurement depths are different at both sites: 6–14 cm in Tumbarumba and 15–46 cm in Daly.

The slopes of the linear regression line between simulated and observed soil water contents are 1.08 in Daly (Figure 3(b)) and 1.04 in Tumbarumba (Figure 3(d)), both of which are very close to 1.0. As shown in Table 2, the MAE and RMSE for soil water content are 0.02 and 0.03 $\text{cm}^{-3} \text{cm}^{-3}$ in Daly, and 0.04 and 0.05 $\text{cm}^{-3} \text{cm}^{-3}$ in Tumbarumba. The NSE is 0.74 in Daly and 0.61 in Tumbarumba. Therefore, the simulated soil water contents agree well with the measured soil water contents at both sites.

The results are not consistent with those of Renzullo *et al.* (2014), which showed that soil water content in Tumbarumba does not have a high correlation between the CosmOz measured soil moisture data and the simulation results with the

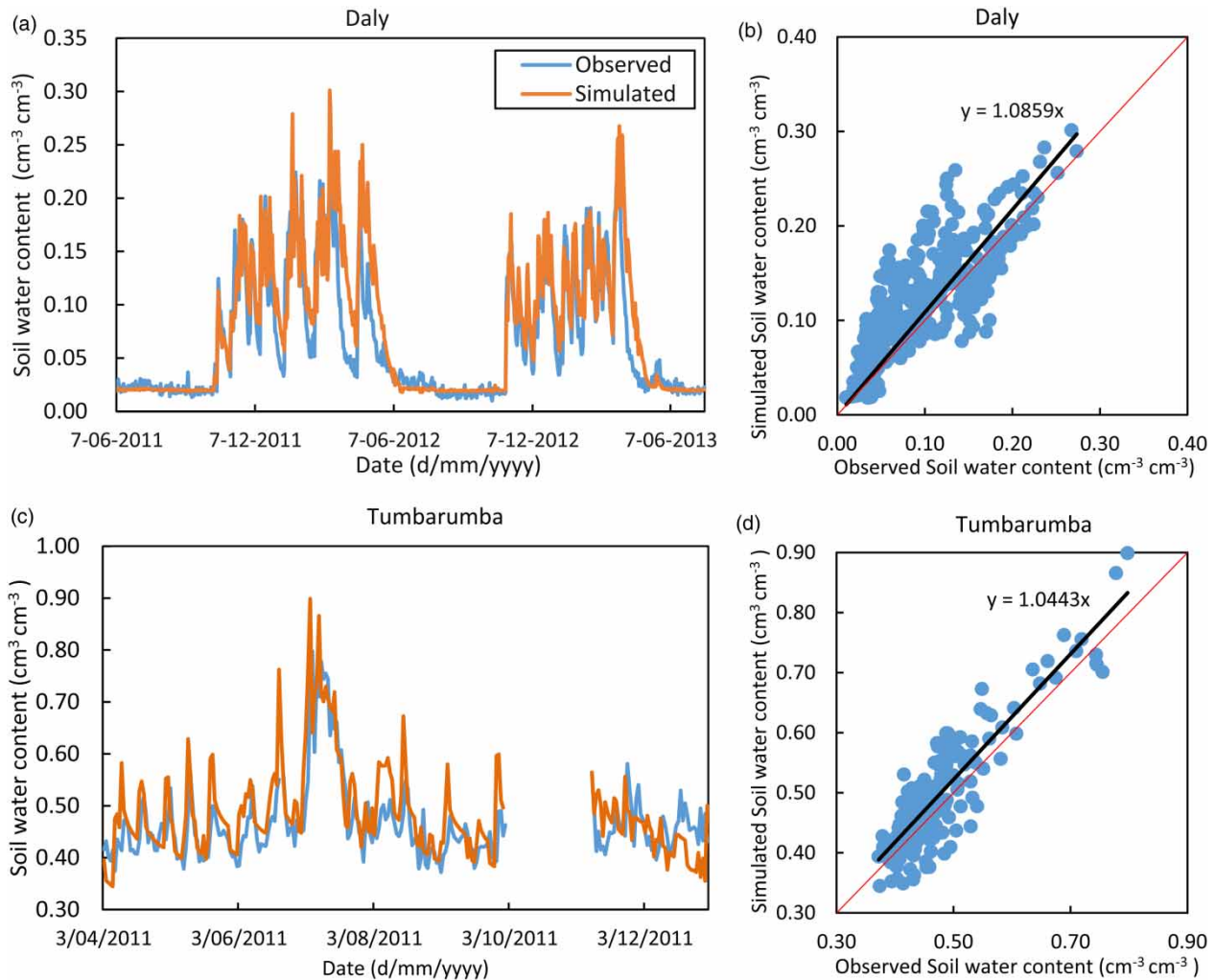


Figure 3 | Comparison of simulated and observed soil water contents at the Daly (a, b) and Tumbarumba (c, d) sites.

Table 2 | Statistics for comparison between observed and simulated soil water content and ET at the Daly and Tumbarumba sites

Study site	Soil water content			ET		
	MAE ($\text{cm}^3 \text{cm}^{-3}$)	RMSE ($\text{cm}^3 \text{cm}^{-3}$)	NSE	MAE (mm day^{-1})	RMSE (mm day^{-1})	NSE
Daly	0.02	0.03	0.74	0.55	0.73	0.60
Tumbarumba	0.04	0.05	0.61	0.36	0.52	0.78

AWRA-L model. The reason may be that the effective measurement depth by CosmOz in Tumbarumba is too shallow (6–14 cm) (Hawdon *et al.* 2014) and the effective measurement depth is strongly dependent on soil moisture (~75 cm for dry soils and ~12 cm for wet soils) (Bogena *et al.* 2013). All the evaluation indices for soil water content are higher at both sites, so we considered that the WAVES model is accurate for simulating the soil water content in Daly and Tumbarumba.

Evapotranspiration

Figure 4 shows the comparison of the simulated ET by WAVES with the observed ET by the eddy covariance method in Daly and Tumbarumba. Figure 4(a) and 4(c) presents that the simulated ET have the same seasonal patterns as the observed one, although there are some days when the WAVES model did not match the measurements. Figure 4(a) and 4(c) shows that the

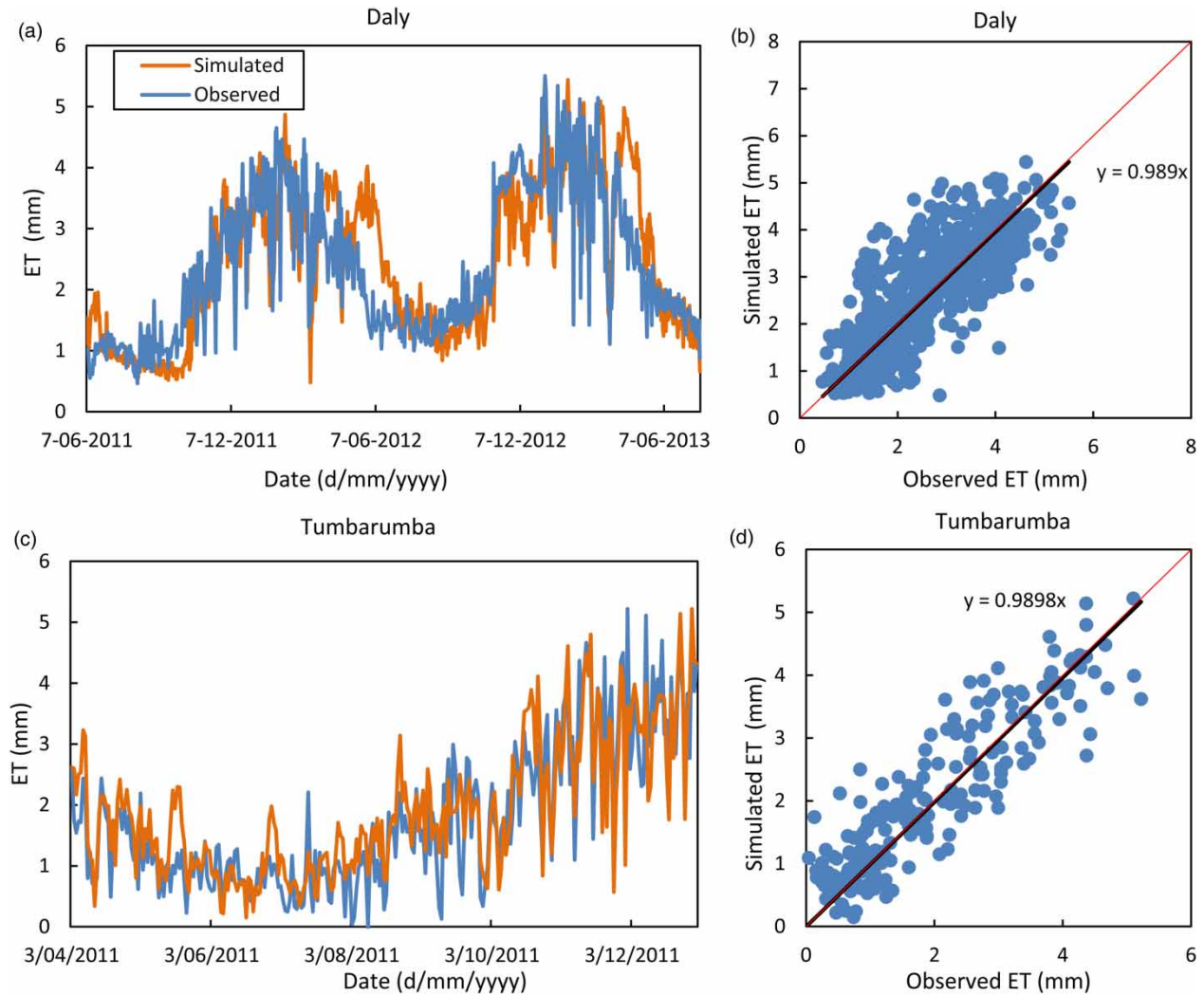


Figure 4 | Comparison of simulated and observed ET at the Daly (a, b) and Tumbarumba (c, d) sites.

ET has a strong seasonal variation at both sites, which has higher values from November to May than those from June to October. The average ET values are larger in Daly than those in Tumbarumba.

Figure 4(b) and 4(d) indicates that both the simulated and observed ET showed a good agreement at both sites. The slopes of the regression line between simulated and measured through the origin are both 0.99, very close to 1.0 (Figure 4(b) and 4(d)). Similarly, the MAE and RMSE for ET are 0.55 and 0.73 mm day⁻¹ in Daly and 0.36 and 0.52 mm day⁻¹ in Tumbarumba (Table 2). The NSE is 0.60 in Daly and 0.78 in Tumbarumba. These indices indicated that the WAVES model can accurately capture the daily variations ET at both sites.

Soil water content and ET were accurately simulated at both sites against the measurements, indicating that the WAVES model is capable of accurately simulating the soil water content and plant water use in the two ecosystems. The success of the WAVES model is consistent with the fact that it coupled the atmosphere, vegetation and soil systems together by modelling the interactions and feedbacks between them and strikes the balance between the complexity of the model, the ease of use and the accuracy of the model (Zhang & Dawes 1998).

Climate variability over the period of 1950–2015

It is known that potential evapotranspiration (PET) and precipitation are generally considered as the main drivers of water balance at the catchment scale. The analysis of trends and variabilities in PET and precipitation can help us to understand

how they affect the water balance components. The inter-annual variations of precipitation and PET during the period of 1950–2015 are shown in Figure 5, and the corresponding monthly mean climatic variations are presented in Figure 6.

Figure 5 shows that in Daly, the annual precipitation has a significant increasing trend ($5.496 \text{ mm year}^{-1}$) and PET has a small decreasing trend ($-0.043 \text{ mm year}^{-1}$) together with the decreasing trend of temperature ($-0.002 \text{ }^{\circ}\text{C year}^{-1}$) over the period of 1950–2015. In Tumbarumba, annual precipitation decreased at the rate of $-1.8 \text{ mm year}^{-1}$, and PET significantly increased at the rate of $1.311 \text{ mm year}^{-1}$ with the increasing temperature at the rate of $0.002 \text{ }^{\circ}\text{C year}^{-1}$ from 1950 to 2015 (Figure 5).

Figure 6 indicates that the mean annual precipitation and the annual PET have different seasonal distributions in Daly and Tumbarumba. In Daly, the average monthly precipitation is 220 mm from November to March, while it is $<50 \text{ mm}$ from April to October (Figure 6(a)). The lowest total monthly PET occurred in February (120 mm) due to high summer rainfall, high

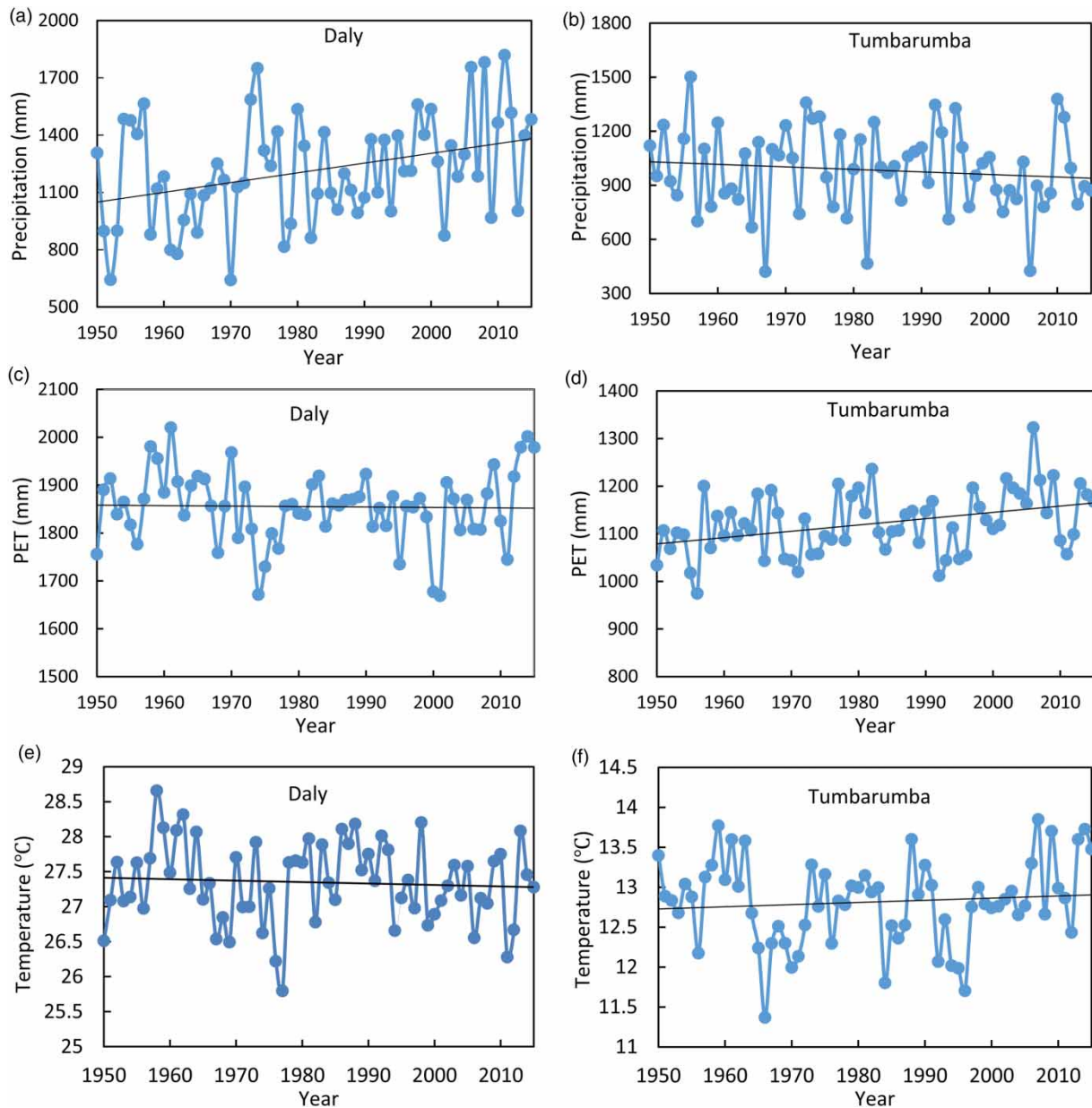


Figure 5 | Inter-annual variability of precipitation (a, b), potential evapotranspiration (PET) (c, d) and temperature (e, f) over the period of 1950–2015 at the Daly (a, c and e) and Tumbarumba (b, d and f) sites. The black lines indicate the linear trends in precipitation, PET and temperature.

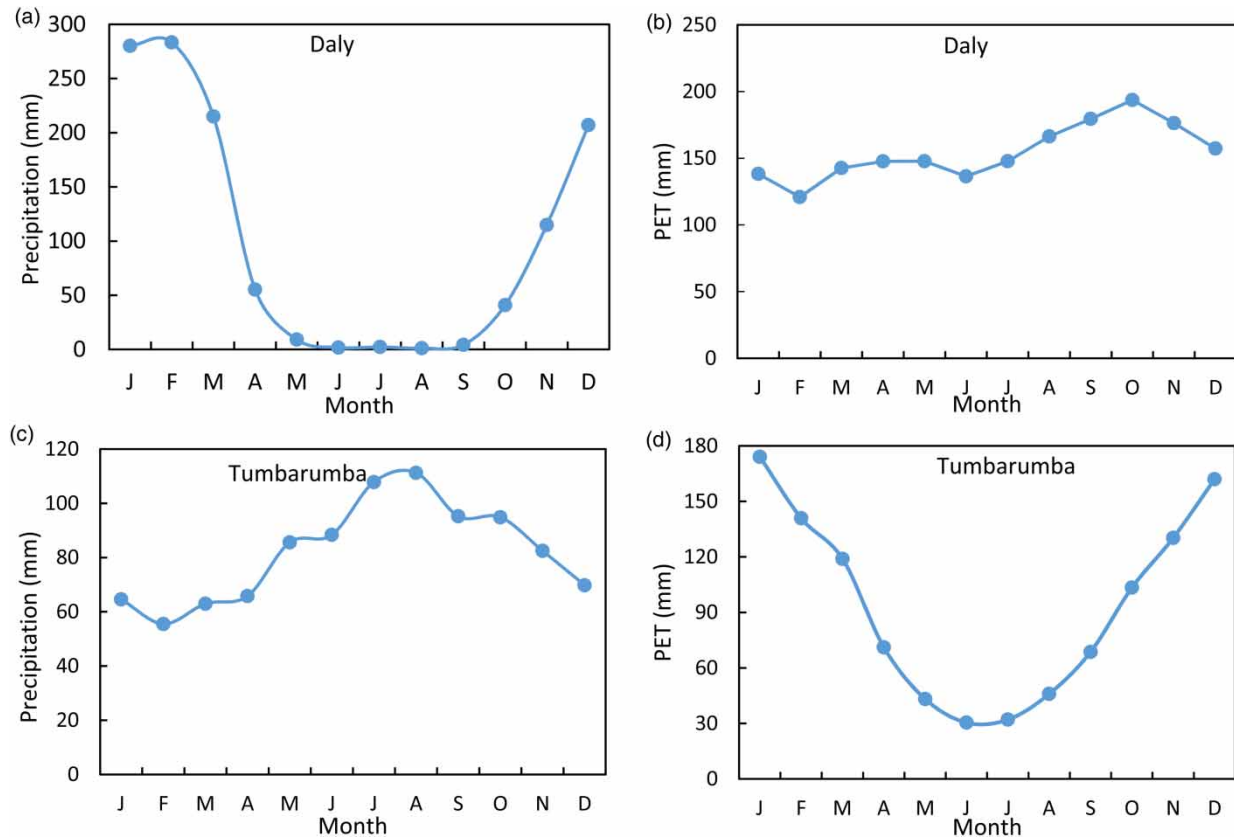


Figure 6 | Monthly mean precipitation (a, c) and potential evapotranspiration (PET) (b, d) during 1950–2015 at the Daly (a, b) and Tumbarumba (c, d) sites.

temperature, low radiation and low vapor pressure deficit (VPD) (Figure 6(b)). After the minimum PET values in February, PET has a linear increase to the maximum monthly value in October (193 mm) (Figure 6(b)). The PET is a common method to estimate ET for the ecohydrological study. Therefore, the tropical climatic conditions are the main factors which lead to the large variation of soil water content and ET in Daly.

In Tumbarumba (Figure 6(c) and 6(d)), the monthly precipitation and PET are out of the phase. The mean monthly precipitation peaks from June to September (>80 mm), while the minimum mean monthly PET occurs from May to August (<45 mm) and the maximum one is from November to February (>140 mm). Since the precipitation is distributed more evenly throughout the year in Tumbarumba, correspondingly the monthly variation of soil water content is relatively low.

Annual trends and variability of water balance components over the past 60 years

Evapotranspiration

Figure 7 presents the simulated annual total ET during 1950–2015 in Daly and Tumbarumba using optimized model parameter values. The MK test and Sen's slope values of water balance components during 1950–2015 in Daly and Tumbarumba are shown in Table 3. ET includes the evaporation from interception (canopy and litter layer), canopy transpiration and soil evaporation. The ET ranged from 511 to 1,070 mm in Daly, and 539 to 992 mm in Tumbarumba. The annual ET has a clear decreasing trend ($-1.988 \text{ mm year}^{-1}$) in Daly, while it presented a slight decreasing trend ($-0.381 \text{ mm year}^{-1}$) in Tumbarumba during the period of 1950–2015. Therefore, the annual total ET has the same tendency at both sites during the study periods, although the tendency variation is different.

Runoff

Figure 8 shows the simulated annual runoff during 1950–2015 in Daly and Tumbarumba. The annual runoff varied between 192 and 1,304 mm in Daly, and 31 and 401 mm in Tumbarumba. The annual runoff increased significantly ($5.870 \text{ mm year}^{-1}$)

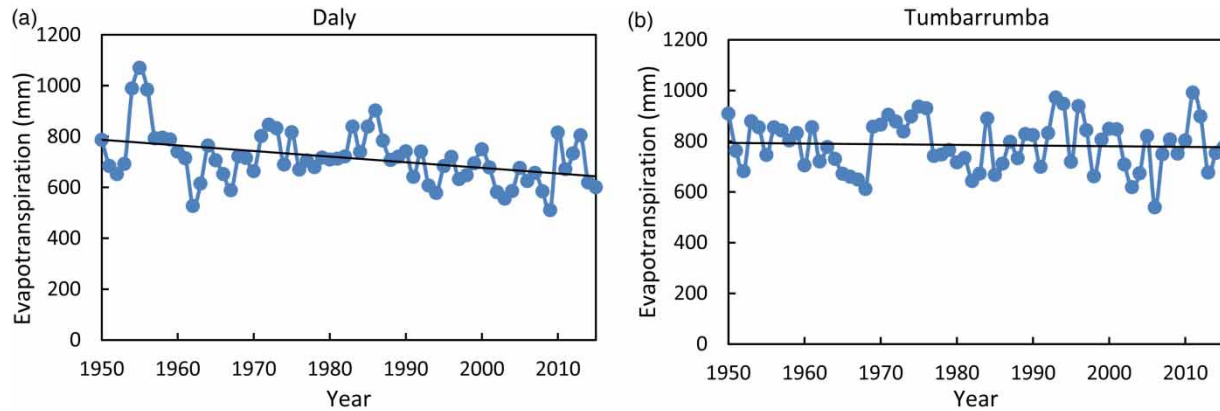


Figure 7 | Simulated annual total ET at the Daly (a) and Tumbarumba (b) sites during 1950–2015. The black lines show the trends in the simulated ET.

Table 3 | Summary of estimated trend for key variables using the Mann–Kendall test

Items/sites	Daly			Tumbarumba			Slope (Q)	
	Test Z	Significance	Slope (Q)	Test Z	Significance	Slope (Q)		
Precipitation	2.83	**	mm year ⁻¹	5.496	–1.08	ns	mm year ⁻¹	–1.800
T_{mean}	–0.42	ns	°C year ⁻¹	–0.002	0.48	ns	°C year ⁻¹	0.002
PET	–0.12	ns	mm year ⁻¹	–0.043	3.15	**	mm year ⁻¹	1.311
ET	–3.01	**	mm year ⁻¹	–1.988	–0.51	ns	mm year ⁻¹	–0.381
Runoff	3.59	***	mm year ⁻¹	5.870	–1.42	ns	mm year ⁻¹	–0.886
LAI	–3.71	***		–1.855	2.67	**		0.868
Soil water content 0.1 m	2.74	**	cm ³ m ⁻³ year ⁻¹	0.0002	–0.91	ns	cm ³ cm ⁻³ year ⁻¹	–0.0001
0.5 m	3.38	***	cm ³ cm ⁻³ year ⁻¹	0.0004	–1.10	ns	cm ³ cm ⁻³ year ⁻¹	–0.0001
1.0 m	3.97	***	cm ³ cm ⁻³ year ⁻¹	0.0008	–2.08	*	cm ³ cm ⁻³ year ⁻¹	–0.0002
1.5 m (3.0 m)	4.32	***	cm ³ cm ⁻³ year ⁻¹	0.0008	–3.45	***	cm ³ cm ⁻³ year ⁻¹	–0.0007

Notes: Soil depth of 1.5 m for Daly and soil depth of 3.0 m for Tumbarumba. ***, ** and * indicate significance levels of 0.001, 0.01 and 0.05; ns means significance level exceeds 0.05.

in Daly, while there was a decreasing trend (–0.886 mm year⁻¹) in Tumbarumba during the period of 1950–2015. Therefore, the annual runoff in Daly varies more obviously than that in Tumbarumba.

Soil water content

Figure 9 displays the mean annual soil water content at different depths during 1950–2015 in Daly. Soil water content in Daly is in the ranges of 0.05–0.11, 0.04–0.13, 0.04–0.21 and 0.04–0.27 cm³ cm⁻³ at the soil depths of 0.1, 0.5, 1.0 and 1.5 m, respectively. The soil water content shows increasing trends at all depths, while the slope of the trend increases with the soil depth (Table 3). Soil water content increases in wet years and decreases in dry years, but the amplitudes of variation are different at different depths.

Figure 10 illustrates the mean annual soil water content at different depths during 1950–2015 in Tumbarumba. The soil water content in Tumbarumba is within the ranges of 0.39–0.47, 0.37–0.47, 0.35–0.46 and 0.39–0.51 cm³ cm⁻³ at 0.1, 0.5, 1.0 and 3.0 m, respectively. The soil water content decreases with time, and the slope of the trend increases with soil depths (Table 3). For the shallow soil, the soil water content showed similar patterns (Figure 10(a), 10(b) and 10(c)). For the deep soil (Figure 10(d)), the soil moisture content exhibited the different pattern. The soil water content at 3.0 m in the later years is much higher than that in the surface soil at the corresponding time.

Climate variables have exhibited different trends and variations in the two ecosystems. In Daly with the tropical climate, ET is mainly controlled by the plant-available water and canopy resistance for savanna. In Tumbarumba with an oceanic climate,

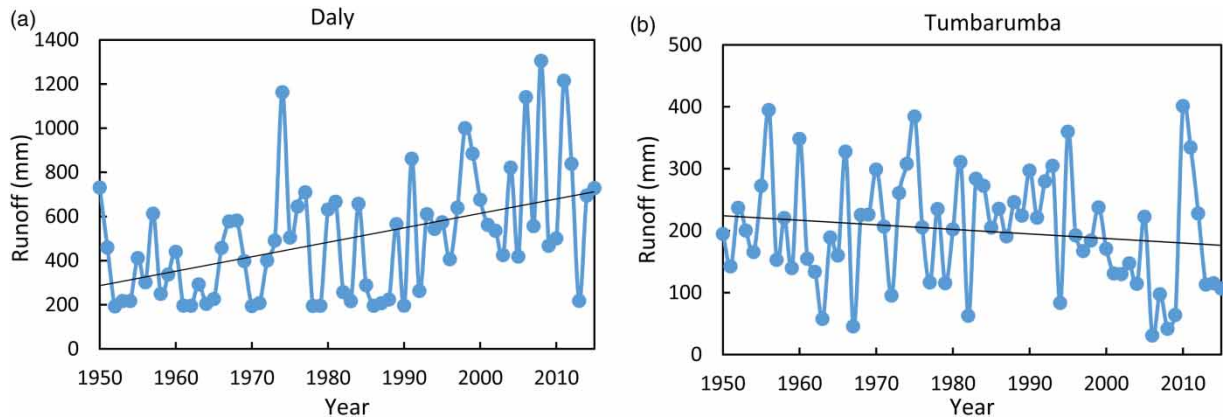


Figure 8 | Simulated annual runoff at the Daly (a) and Tumbarumba (b) sites during 1950–2015. The black lines show the trends in the simulated runoff.

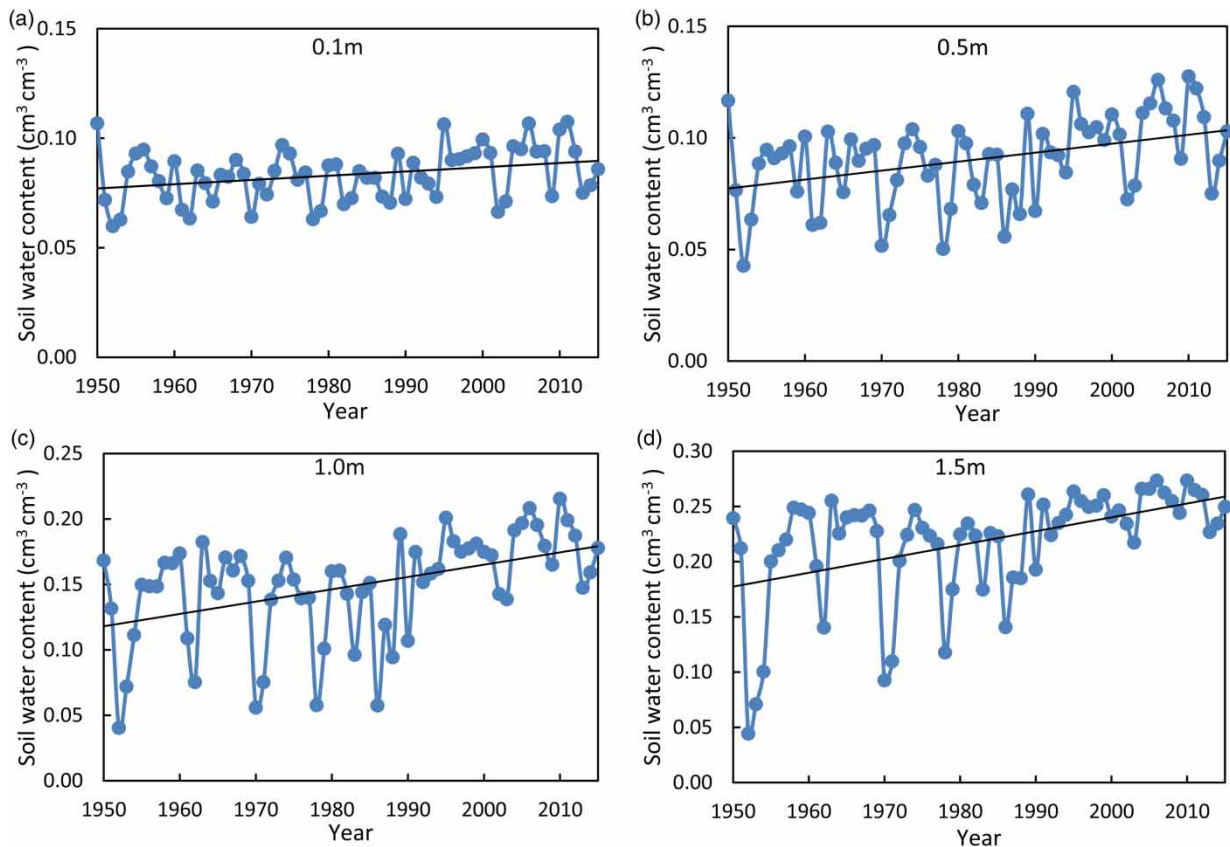


Figure 9 | Mean annual soil water content at the depths of 0.1 m (a), 0.5 m (b), 1.0 m (c) and 1.5 m (d) over the studied period of 1950–2015 at the Daly site. The black lines show the trends in the simulated soil water content.

ET is dominated by the net radiation, leaf area and advection. Annual runoff showed an increasing trend in Daly and decreasing in Tumbarumba over the periods of 1950–2015. In Daly, rainfall is highly uneven throughout the year with distinct wet and dry seasons (Figure 5(a)), and it increased at the rate of $5.496 \text{ mm year}^{-1}$ during 1950–2015. The increasing rainfall in the wet season contributed to increased runoff (Table 3). The soil cannot supply enough water for vegetation when the dry season comes; therefore, the ET and the LAI decrease.

In Tumbarumba, the precipitation is distributed relatively evenly all over the year (Figure 5(c)) and decreased at the rate of $-1.8 \text{ mm year}^{-1}$ during 1950–2015, which leads to a little reduction of soil water content in the surface soil but more

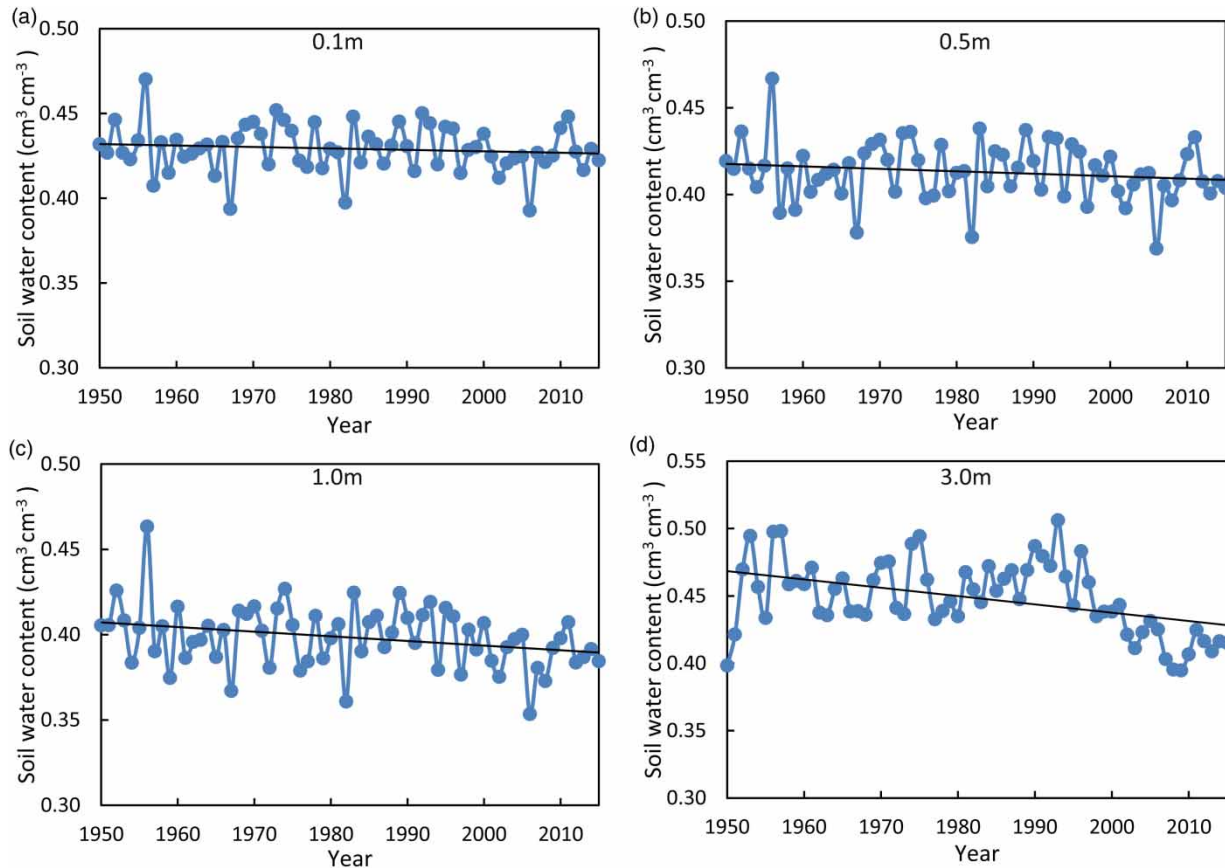


Figure 10 | Mean annual soil water content at depths of 0.1 m (a), 0.5 m (b), 1.0 m (c) and 3.0 m (d) over the studied period of 1950–2015 at the Tumbarumba site. The black lines show the trends in the simulated soil water content.

significant reduction in deep soil. The runoff trends during 1950–2015 are consistent with the annual rainfall trends, suggesting that rainfall is a key factor controlling water balance dynamics of the forest ecosystem (Leuning *et al.* 2005). Soil water content has a similar trend to precipitation for both ecosystems (Table 3). The variation of soil water content is influenced by climate and vegetation type, especially for the topsoil layer. With the uneven distribution of rainfall between months, the soil water content is low in dry seasons and relatively high in wet seasons at the topsoil. The vegetation in Daly is savanna, and it has less effect on deep soil water content since its root depth is <150 cm. The variation slope of mean soil water content in Daly increased with the increasing soil depths of 0.1, 0.5 and 1.0 m, but decreased at the soil depth of 1.5 m, which also indicated that the factors of climate and vegetation have less influence on the deep soil than on the topsoil.

In Tumbarumba, the annual soil water content has an increasing trend at the soil depths of 0.1, 0.5 and 1.0 m, but lightly decreasing trend at the soil depth of 3 m. Since Tumbarumba has an oceanic climate, the precipitation is evenly dispersed throughout the year (Figure 6(c)), which leads to the relatively stable variation of mean annual soil water content at different soil depths (Figure 10). Meanwhile, the vegetation is eucalyptus which has deep root depth and can increase the variation of soil water content at the deep soil depth. The litter layer of eucalyptus is also an important factor to influence the soil water content. The variation slope of soil water content in Tumbarumba increased with the increasing soil depths of 0.1, 0.5, 1.0 and 3.0 m, but it is much lower than that in Daly. This is closely related to the precipitation distribution during the year (Figure 6(c)).

In addition, the vegetation types also have effects on the variation of water balance components along with climate change (Gao *et al.* 2015). The vegetation type is eucalyptus forest in Tumbarumba and woodland savanna in Daly. Both ecosystems have different canopy, litter layer and root depth, all of which will influence the distribution and the trends of water balance components. The eucalyptus has a much larger canopy than the savanna, which results in different precipitation interception,

ET and runoff. Although the litter layer can increase the soil water storage and decrease surface flow, different depths of the litter layer will have different effects on two ecosystems. Since the root depth of eucalyptus is much deeper than that of savanna, the soil water content in Tumbarumba varies more conspicuously at deep soil than that in Daly. Meanwhile, the eucalyptus in Tumbarumba can extract water from the deeper soil because of the deep root system when the precipitation is decreasing.

CONCLUSIONS

In this study, climate variability impacts on water balance components (ET, runoff and soil water content) were studied for two different ecosystems in Australia using an ecohydrological model such as WAVES. The soil moisture and ET measurements obtained from the CosmOz and OzFlux networks were used to calibrate the WAVES model, and the calibrated model was used to investigate trends and variability in water balance components over the period of 1950–2015 at these two ecosystems.

The following conclusions can be drawn from this study:

- The WAVES model can be accurately calibrated against measured soil water content and ET in Daly and Tumbarumba.
- The precipitation and temperature at two sites showed different variabilities and trends over the period of 1950–2015, which lead to the changes of ET and soil water content at different soil depths. Precipitation has a direct effect on soil water content; when precipitation increases, the soil water content increases and it changes more conspicuously at deep soil. The variations of ET and runoff are dependent not only on precipitation but also on the plant growth conditions for different vegetation types.

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DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

REFERENCES

- Beringer, J., Hutley, L. B., Tapper, N. J. & Cernusak, L. A. 2007 Savanna fires and their impact on net ecosystem productivity in North Australia. *Global Change Biology* **13**, 990–1004.
- Bogena, H. R., Huisman, J. A., Baatz, R., Hendricks, F. H. J. & Vereecken, H. 2013 Accuracy of the cosmic-ray soil water content probe in humid forest ecosystems: the worst case scenario. *Water Resources Research* **49**, 5778–5791.
- Cernusak, L. A., Hutley, L. B., Beringer, J., Holtum, J. A. & Turner, B. L. 2011 Photosynthetic physiology of eucalypts along a sub-continental rainfall gradient in northern Australia. *Agricultural and Forest Meteorology* **151**, 1462–1470.
- Chen, C., Cleverly, J., Zhang, L., Yu, Q. & Eamus, D. 2016 Modelling seasonal and inter-annual variations in carbon and water fluxes in an arid-zone acacia savanna woodland, 1981–2012. *Ecosystems* **19**, 625–644.
- Cheng, L., Zhang, L., Wang, Y., Yu, Q. & Eamus, D. 2014a Quantifying the effects of elevated CO₂ on water budgets by combining FACE data with an ecohydrological model. *Ecohydrology* **7**, 1574–1588.
- Cheng, L., Zhang, L., Wang, Y., Yu, Q., Eamus, D. & O'Grady, A. 2014b Impacts of elevated CO₂, climate change and their interactions on water budgets in four different catchments in Australia. *Journal of Hydrology* **519**, 1350–1361.
- Crosbie, R. S., McCallum, J. L., Walker, G. R. & Chiew, F. H. S. 2010 Modelling climate-change impacts on groundwater recharge in the Murray-Darling Basin, Australia. *Hydrogeology Journal* **18** (7), 1639–1656.
- Dawes, W. R. & Short, D. L. 1993 *The Efficient Numerical Solution of Differential Equations for Coupled Water and Solute Dynamics: The WAVES Model*. CSIRO Division of Water Resources, Canberra.
- Dawes, W. R., Zhang, L. & Dyce, P. 1998 *WAVES V3.5 User Manual*. CSIRO Land and Water, Canberra, Australia.
- Gao, Z., Zhang, L., Cheng, L., Zhang, X., Cowan, T., Cai, W. & Brutsaert, W. 2015 Groundwater storage trends in the Loess Plateau of China estimated from streamflow records. *Journal of Hydrology* **530**, 281–290.
- Glenn, E. P., Doody, T. M., Guerschman, J. P., Huete, A. R., King, E. A., McVicar, T. R., Van Dijk, A. I. J. M., Van Niel, T. G., Yebra, M. & Zhang, Y. 2011 Actual evapotranspiration estimation by ground and remote sensing methods: the Australian experience. *Hydrological Processes* **25**, 4103–4116.
- Hawdon, A., McJannet, D. & Wallace, J. 2014 Calibration and correction procedures for cosmic-ray neutron soil moisture probes located across Australia. *Water Resources Research* **50**, 5029–5043.

- Kang, S., Zhang, L., Liang, Y. & Dawes, W. 2003 Simulation of winter wheat yield and water use efficiency in the Loess Plateau of China using WAVES. *Agricultural Systems* **78** (3), 355–367.
- Kendall, M. G. 1975 *Rank Correlation Measures*. Charles Griffin, London.
- Kray, J. A., Cooper, D. J. & Sanderson, J. S. 2012 Groundwater use by native plants in response to changes in precipitation in an intermountain basin. *Journal of Arid Environments* **83**, 25–34.
- Leuning, R., Cleugh, H. A., Zegelin, S. J. & Hughes, D. 2005 Carbon and water fluxes over a temperate Eucalyptus forest and a tropical wet/dry savanna in Australia: measurements and comparison with MODIS remote sensing estimates. *Agricultural & Forestry Meteorology* **129**, 151–173.
- Leuning, R., Zhang, Y. Q., Rajaud, A., Cleugh, H. & Tu, K. 2008 A simple surface conductance model to estimate regional evaporation using MODIS leaf area index and the Penman-Monteith equation. *Water Resources Research* **44**, W10419.
- Mann, H. B. 1945 Nonparametric test against trend. *Econometrica* **13** (3), 245–259.
- Nash, J. E. & Sutcliffe, J. V. 1970 River flow forecasting through conceptual models part I – a discussion of principles. *Journal of Hydrology* **10**, 282–290.
- Prior, L. D. & Eamus, D. 1999 Seasonal changes in leaf water characteristics of *Eucalyptus tetrodonta* and *Terminalia ferdinandiana* saplings in a northern Australian savanna. *Australian Journal of Botany* **47**, 587–599.
- Renzullo, L. J., van Dijk, A. I. J. M., Perraud, J. M., Collins, D., Henderson, B., Jin, H., Smith, A. B. & McJannet, D. L. 2014 Continental satellite soil moisture data assimilation improves root-zone moisture analysis for water resources assessment. *Journal of Hydrology* **519**, 2747–2762.
- Salmi, T., Maata, A., Antilla, P., Ruoho-Airola, T. & Amnell, T. 2002 *Detecting Trends of Annual Values of Atmospheric Pollutants by the Mann-Kendall Test and Sen's Slope Estimates – The Excel Template Application Makesens*. Finnish Meteorological Institute, Helsinki, Finland.
- Tian, F., Feng, X. M., Zhang, L., Fu, B. J., Wang, S., Lv, Y. & Wang, P. 2017 Effects of revegetation on soil moisture under different precipitation gradients in the Loess Plateau, China. *Hydrological Research* **48** (5), 1378–1390.
- Wang, S., Fu, B., Gao, G., Liu, Y. & Zhou, J. 2013 Responses of soil moisture in different land cover types to rainfall events in a re-vegetation catchment area of the Loess Plateau, China. *Catena* **101**, 122–128.
- Ye, L., Cheng, L., Liu, P., Liu, D., Zhang, L., Qin, S. & Xia, J. 2021 Management of vegetative land for more water yield under future climate conditions in the over-utilized water resources regions: a case study in the Xiong'an New area. *Journal of Hydrology* **600**, 126563.
- Zhang, L. & Dawes, W. R. 1998 WAVES: An Integrated Energy and Water Balance Model. CSIRO Land and Water, Canberra, Australia.
- Zhang, L., Dawes, W. R. & Hatton, T. J. 1996 Modelling hydrologic processes using a biophysically based model application of WAVES to FIFE and HAPEX-MOBILHY. *Journal of Hydrology* **185**, 147–169.
- Zhang, L., Dawes, W. R., Slavich, P. G., Meyer, W. S., Thorburn, P. J., Smith, D. J. & Walker, G. R. 1999 Growth and around water uptake responses of lucerne to changes in groundwater levels and salinity: lysimeter, isotope and modelling studies. *Agricultural Water Management* **39**, 265–282.
- Zhu, X., Shao, M., Jia, X., Huang, L., Zhu, J. & Zhang, Y. 2017 Application of temporal stability analysis in depth-scaling estimated soil water content by cosmic-ray neutron probe on the northern Tibetan Plateau. *Journal of Hydrology* **546**, 299–308.
- Zreda, M., Shuttleworth, W. J., Zeng, X., Zueck, C., Desilets, D., Franz, T. E. & Rosolem, R. 2012 COSMOS: the cosmic-ray soil moisture observing system. *Hydrology Earth System Sciences Discussions* **16**, 4079–4099.

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